

Persistent Use of Overtime among Fixed-Term (Contrata) Staff of the Central Metropolitan Health Service

Data Preparation and Exploratory Analysis

C1 · From the W1 Project to Defensible Evidence

Group 01

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Abstract

This report develops the line selected in W1 (Candidate B): the persistent use of overtime among fixed-term (contrata) staff of the Central Metropolitan Health Service (Servicio de Salud Metropolitano Central, SSMC). The proposed user is the SSMC's people management and development units, and the decision the analysis seeks to support is which staff categories and positions to review first for staffing. The source is the monthly payroll of contrata staff published under Active Transparency (Transparencia Activa) between September 2024 and August 2026 (24 files, 44,777 records). The records were aggregated to 43,120 person-months and profiled, and problematic cases (atypical month, impossible values and extreme values) were flagged instead of removed.

Of the 1,776 people present for at least 12 months, 364 (20.5%) received overtime in at least 75% of their months. The rate stays between 19.0% and 20.5% when flagged cases are excluded, and ranges from 28.2% to 14.0% with thresholds of 50% and 90%, without changing which staff categories have the highest and lowest persistence. Persistence is concentrated among auxiliary (55.4%) and administrative staff (44.8%), and within them among drivers, administrative support and secretaries; among professionals, in clinical care (25.9%). People who receive overtime more frequently also tend to have a larger share of it in their gross pay (Spearman 0.58), also within each staff category except physicians; auxiliary staff have the highest median share (24.8%, versus 14.6% overall). Monthly frequency fell from 28.4% to 26.4% between the two years of the period, in all five staff categories.

The payroll records payments and does not report authorized staffing, vacancies, attendance or facility. The analysis therefore makes it possible to prioritize where to review staffing, but not to demonstrate a staff shortage or explain the cause of persistence. As a continuation, we propose a supervised task to anticipate persistent use in the following year and a dashboard for people management.

Keywords: overtime; Active Transparency; people management; health services; data quality; exploratory analysis.

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Introduction

In W1, the team explored two lines of inquiry on overtime among SSMC contrata staff: Candidate A, focused on payments above the legal limit of 40 daytime hours and on atypical cases, and Candidate B, focused on persistent overtime use as a possible signal of a staffing gap. The team provisionally preferred Candidate B. This report develops it into a defensible analysis: it bounds the problem and the decision, formalizes the metrics, documents data preparation and quality, presents the exploratory data analysis (EDA) and concludes within what the evidence allows.

Section 1 presents the problem, its documentary support and its evolution since W1. Section 2 applies the seven steps from Class 03 and defines the metrics. Section 3 describes the data, their preparation and the Five Cs. Section 4 presents the EDA results. Section 5 sets out the conclusions and limitations, and Section 6 the continuation toward machine learning and a dashboard. Section 7 contains the Process record, followed by the references and appendices. The reproducible analysis is in `analysis/EDA.ipynb`; references to the notebook indicate the heading of the corresponding section.

1. Selected problem and evolution since W1

1.1 Problem, population, place, period and proposed use

Problem. The persistent, month-to-month use of overtime to cover the functions of SSMC contrata staff. When a person receives overtime almost every month, it stops being an occasional resource for tasks that cannot be postponed and becomes a routine way of covering a function, which may indicate that staffing for that function is insufficient.

Population, place and period. Contrata staff of the SSMC, a service under the Undersecretariat of Healthcare Networks of the Ministry of Health (MINSAL), whose network covers the communes of Santiago, Estación Central, Cerrillos, Maipú and Pedro Aguirre Cerda and also includes SAMU Metropolitano, the prehospital emergency service (SSMC, n.d.-a, n.d.-c). The period covers the payment months from September 2024 to August 2026. The source does not include permanent (planta) staff, fee-based (honorarios) staff or staff governed by the Labor Code.

Proposed use (team proposal). To enable the SSMC's people management and development units to prioritize which staff categories and positions to review first for staffing, rather than sustaining recurring overtime payments. The analysis does not seek to detect fraud or evaluate individual employees; results are presented in aggregate.

1.2 Documentary support and relation to health policy

Legal framework. Law 18.834 allows extraordinary work to be ordered for tasks that cannot be postponed, compensated with time off or, if that is not possible, with a pay surcharge (Law 18.834, arts. 66 and 69, cited in Contraloría General de la República [CGR], 2026). The surcharge is 25% for daytime hours and 50% for night hours when they cannot be compensated with time off (FALMED, n.d.-b). In health services, overtime is not permanent pay and is not paid during annual leave, medical leave or permits (DFL No. 1 of 2005, art. 97, cited in CGR, 2026). Since July 19, 2025, the Comptroller General has specified that overtime may only be paid for work actually performed (Ruling E121666 of 2025, cited in Diario Constitucional, 2025).

Evidence that the problem is relevant. The Comptroller General's Final Report No. 70 of 2026 on Hospital San José de Victoria showed that overtime use in a health facility can be widespread: 1,000

employees received it over 18 months, for a total of CLP 2,086,479,906 (CGR, 2026). The service's management explained that it responded to the need to ensure continuity of care and cover absences in a 24-hour shift system (BioBioChile, 2026; Las Noticias de Malleco, 2026), which suggests that overtime can operate as a mechanism to compensate for staffing gaps. The international literature associates long working hours and overtime with fatigue, errors and risks for staff (Clendon & Gibbons, 2015; Scott-Marshall, 2024), and the Canadian Institute for Health Information uses overtime as a hospital performance indicator because excessive overtime is associated with lower staff well-being, higher costs and lower quality and safety of care (CIHI, 2026). This evidence comes from other contexts and is used to justify the concern, not to estimate effects in the SSMC.

Health policy: objective, target and indicator. The National Health Strategy for the Health Objectives to 2030 (MINSAL, 2022, March 2022 edition) identifies, among the network's problems, inequity in the distribution of human resources and shortcomings in selection and recruitment; its updated version (MINSAL, 2026, published in April 2026) redefined topics, objectives and indicators. Neither version sets a numerical target for overtime. This work therefore distinguishes three levels: the **policy objective** (adequate and well-distributed staffing), which the payroll does not measure; the **numerical target**, which does not exist for this topic and which the team does not invent; and the **indicator the data can actually measure**, which is the persistent use of paid overtime, treated as a possible symptom of that problem and not as its direct measurement.

1.3 Alternatives considered in W1 and selection

Table 1

Comparison of the W1 candidates and reason for selection

Dimension	Candidate A: above the limit and atypical cases	Candidate B: persistent use (selected)
Question	What share of staff exceeds 40 paid daytime hours, and where are atypical night hours concentrated?	Which staff categories and positions receive overtime persistently, and what share of their pay does it represent?
User and decision	Internal audit: which authorizations to review first	People management: where to review staffing first
Signal in the data (W1)	182 person-months above the limit; 99 in January 2025	364 of 1,776 people (20.5%) with persistent use
Sensitivity to data quality	The main result depends on one atypical month	The December 2025 staff-category error is neutralized by using each person's most frequent category
Main limitation	No attendance data or authorized exceptions	No authorized staffing or vacancy data: describes use, does not demonstrate a gap

Note. Own elaboration based on the W1 report (Sections 3.1 and 3.2).

Candidate B was selected for three reasons documented in W1: its signal is broad and stable, it is less sensitive to the quality problems detected, and it uses the full 24-month series, which provides a better basis for project continuity. Candidate A's 40-daytime-hour limit is retained as contextual background.

1.4 Feedback and changes since W1

W1 posed four questions for instructor feedback (Section 3.3 of W1). **No feedback on those questions was available before this submission.** The team kept the provisional W1 decision (Candidate B) and resolved each pending question with a justified decision, recorded in Table 2 and Section 7.2.

Table 2*Changes since W1 and decisions on the pending questions*

Aspect	In W1	In C1	Reason
Question wording	Persistence as a signal of a staffing gap	Persistent use, without claiming a gap	The source does not report staffing or vacancies (pending question 1)
Person identification	Full name, no RUT (national ID)	Retained; person-month unit with totals check	No other identifier exists (pending question 2)
January and December 2025	December excluded from the analysis by staff category	January flagged, not removed; December resolved with the most frequent staff category	Their effect on the KPIs is assessed (pending question 3)
Joins	Three external joins proposed	Not performed; only validated internal joins	No common key at the person or position-month level (pending question 4)
Metrics	Exploratory percentages	KPIs specified with numerator, denominator, window and exclusions	Class 01: a KPI is a meaningful ratio
Position or function	95 non-normalized values	Normalized to 80 values and assigned per person	Enables segmentation by position
Data quality	Initial checks	Profile, duplicates, missingness mechanism, flags, KPIs before and after, comparability	Classes 02 and 03
Persistence threshold (75%)	Exploratory definition	Tested with 50% and 90%	Verify that the conclusion does not depend on the threshold

Note. Own elaboration.

2. Seven-step map and metric definitions

Table 3 applies the seven steps from Class 03. The user, the decision and the success criteria are team proposals.

Table 3*Seven-step map (Class 03)*

Step	In this project	Gap identified
1. Decision	User: SSMC people management and development units. Decision: which staff categories and positions to review first for staffing. Alternatives: keep paying recurring overtime, create or expand positions, reassign staff across units, or cover with replacement contracts.	The payroll does not report hiring cost, budget or authorized staffing: it allows prioritization, not evaluation of the full decision.
2. Success criteria	Desired: less recurring reliance on overtime without affecting continuity of care. Measurable now: persistence rate, frequency and pay share, as a baseline.	There is no official target; continuity of care is not observed in the payroll.
3. Business questions	Main: which staff categories and positions among contrata staff receive overtime persistently, and what share of their gross pay does it represent? Supporting: does the result hold with other thresholds and without the atypical months? Does it change over time? Is it concentrated in a few people or positions? Is persistence related to pay share?	The period covers 24 months, and the "why" cannot be answered: the payroll does not record the cause.
4. Business process	Authorization → work performed → attendance record → calculation → payroll payment → monthly publication under Active Transparency.	Only the payment is observed; there is no authorization, attendance, medical leave or annual leave data; the payment month may not be the month worked.

Step	In this project	Gap identified
5. Entities, events and states	Entities: person, contract or appointment, staff category, position. Event: monthly payment of daytime or night overtime. States: present or absent in the month's payroll; with or without overtime; persistent or not.	No RUT; facilities, units and patient encounters are not observed.
6. Data representation	Row = contract-month record (44,777). Person-month key = (period, name): 43,120 rows. Person = name (2,430). Coverage: SSMC contrata staff, 24 months. Joins: internal only (aggregation, and most frequent staff category and position).	Amounts and hours stored as text; "-" in bonuses and additional pay; no records of holiday hours; staff category misrecorded in December 2025.
7. Measurement	Definitions in Table 4.	The payroll records payments, not hours worked, and does not measure staffing.

Note. Own elaboration. Notebook, section C1.0.

Unit of observation. Each payroll row is a contract or appointment record in a month, not a person: between 57 and 68 people per month have more than one record (W1, Section 2.2). Records are therefore summed by person and month. A **person-month** represents one person present in a month's payroll, whether or not they received overtime; a person contributes at most 24 person-months. A month in which the person does not appear in the payroll is not counted as zero: it is simply not part of their history.

Table 4
Metric specification

Metric	Formula	Unit and window	Exclusions
Persistence rate (main KPI)	People with overtime in $\geq 75\%$ of their months present / people present ≥ 12 months $\times 100$	% of people; full period; by staff category and position	People with fewer than 12 months (654 of 2,430)
Frequency	Person-months with overtime / person-months $\times 100$	% of person-months; monthly and annual	None
Pay share	Median of (overtime amount / gross pay $\times 100$) among person-months with overtime	% of gross pay; monthly and full period	Person-months with gross pay equal to zero (3 with overtime)

Note. Overtime = daytime + night hours (holiday hours have no records in the 24 months). The 75% threshold and the 12-month minimum are team definitions; their sensitivity is assessed in Section 4.2. There is no official target: the period values are used as a baseline.

Measurement validity. What we want to know is whether certain positions lack staff; what the KPI measures is whether certain people receive overtime payments recurrently. Persistence is compatible with a staffing shortfall but does not demonstrate it. In addition, CIHI's reference indicator measures overtime as a proportion of total hours worked (CIHI, 2026); since the payroll does not report regular hours, the share of overtime pay in gross pay is used as a proxy.

3. Data, preparation and quality

3.1 Source, coverage and procedure

The data come from the SSMC contrata staff payroll on the Transparency Portal (agency code AO006). The 24 monthly files (September 2024 to August 2026) were downloaded in CSV format on September 15, 2026 and are kept unmodified in the data/ folder. They were read in Python (pandas) with a semicolon separator, latin-1 encoding and all columns as text, to make conversions explicit. Amounts were converted to pesos, overtime columns were split into amount and number of hours ("No tiene" [none] was coded as zero, because it indicates the absence of payment rather than missing data) and

dates were converted to date format, with no unconvertible values except the dashes described in Section 3.2 (notebook, sections 1 and 2).

3.2 Profiling

The 22 original columns were profiled: type, missing values, missing rate and unique values (Class 02). **No column has empty cells**, but this does not mean that no data are missing: in this source, missing values are written as "-" and appear in two columns. In additional pay, the 1,825 dashes correspond to all records from January 2025 and to no other month, so missingness depends on the file, not on the person. In bonuses, the 3,065 dashes are concentrated in May 2025 (1,507; 80.4% of the month) and July 2026 (1,479; 79.6%), with smaller counts in December 2024 (70) and November 2025 (9). In those two months, positive bonuses almost disappear (0.0% and 0.1%), whereas in most months they range between 65% and 71%: the dash appears to take the place of unreported positive amounts, so missingness could depend on the value itself (MNAR mechanism), which the source does not allow us to confirm. By contrast, in October 2024, April 2025 and August 2026 bonuses are zero in almost all records without dashes, indicating payments made in specific months: **a recorded zero is not a missing value**. Dashes were kept as missing, without imputation; none of these columns enters the KPIs, which use gross pay, with no missing values (notebook, section 6.1).

Duplicates. There are 27 rows identical to another row in all columns, including the month (54 rows, 11 people). All belong to the physician category, with 11-, 22- and 28-hour schedules, which correspond to hourly positions in that category (Hospital de Linares, n.d.), and none has overtime. They could be two identical appointments or accidental copies; the payroll has no contract identifier to distinguish them. They were retained because they do not change any KPI.

Other checks. The holiday-hours column shows "No tiene" in all 44,777 records; region has a single value; position has 95 values, which drop to 80 after unifying capitalization and the synonym chofer/conductor (driver); 168 people change position during the period and 228 change staff category, so each person is assigned their most frequent position and staff category (only one person has a tie in the staff-category mode). In addition, 207 records with no declared entitlement to overtime show paid hours (W1, Section 2.4).

3.3 Transformations, flags and effect on the population

Records were summed by person and month. Then, instead of deleting rows, **flags** were created (Class 02: an outlier flag is not a deletion rule), whose effect is assessed in Table 6:

- **Atypical month** (1,756 person-months): January 2025, with a different format (dashes in additional pay, 99 of the 182 cases above 40 daytime hours, and 0% overtime in the physician category).
- **Zero pay** (173 person-months, 3 with overtime): pay share cannot be calculated and is treated as structurally missing.
- **Impossible value** (9 person-months): overtime amount greater than the gross pay that includes it (maximum 451.6%).
- **Extreme value** (486 person-months): overtime above the threshold $Q3 + 1.5 \times IQR = 72 + 1.5 \times 57 = 157.5$ hours, calculated among person-months with overtime. Of these, 61.3% (298) are auxiliary staff, who account for only 6.9% of person-months; the median of these cases is 191.5

hours and the maximum 431. They are extreme but possible as payments (for example, hours accumulated from previous months), so they are retained.

Table 5
Flow of the analyzed population and aggregation check

Stage	Units	Check
Payroll rows (contract-month)	44,777	Equal to the sum of the 24 files
Person-months	43,120	Sum of hours, amounts and gross pay equal before and after grouping
Distinct people	2,430	All with staff category and position assigned
People present ≥ 12 months (persistence denominator)	1,776	654 people outside this KPI
Person-months with overtime (frequency numerator)	11,805	27.4% of person-months
Person-months with overtime and pay > 0 (pay-share base)	11,802	3 excluded for zero pay

Note. Notebook, section 6.4. The processed table was saved as data/processed/analitica_persona_mes.csv, with a numeric identifier instead of the name.

Table 6
KPIs before and after excluding flagged cases

Scenario	Person-months	People ≥ 12 months	Persistent	Persistence (%)	Frequency (%)	Pay share, median (%)
Original	43,120	1,776	364	20.5	27.4	14.6
Without January 2025	41,364	1,760	350	19.9	27.3	14.4
Without impossible values	43,111	1,776	363	20.4	27.4	14.6
Without extreme hours	42,634	1,756	340	19.4	26.5	14.1
Without any flag	40,765	1,737	330	19.0	26.6	13.9

Note. Notebook, section 6.4. The original scenario reproduces the W1 values.

The three KPIs change little: persistence between 19.0% and 20.5%, frequency between 26.5% and 27.4% and pay share between 13.9% and 14.6%. The largest change occurs when extreme hours are excluded (−1.1 points), which was expected: removing a month with overtime removes a month that counted toward the person's persistence, even though that payment is not impossible. **The original scenario is kept as the main value** and the range is reported as sensitivity; no flag changes the conclusion.

3.4 The Five Cs

Table 7
Assessment of the Five Cs

C	What was checked	Result and decision
Clean	Duplicates, impossible and extreme values, coded missing values	27 exact duplicates retained (no overtime); 9 impossible and 486 extreme values flagged; dashes kept as missing, not imputed. KPIs vary by less than 1.5 points across scenarios.
Consistent	Stability of definitions and records across the 24 months	Staff category misrecorded in December 2025 (218 changes and 213 back): resolved with the most frequent staff category. January 2025 with a different format: flagged. August 2025 with an anomalous amount per hour: not comparable for pay share (Section 4.3). Position normalized from 95 to 80 values. Date header "dd/mm/aa" with a four-digit year.

C	What was checked	Result and decision
Conformed	Agreement of entities and keys across sources	Not applicable in the strict sense: no external sources are joined, because the payroll does not report RUT or facility, and the joins proposed in W1 (permanent staff, Budget Law, REM statistics) would only be annual or aggregated to the Service level. Internal joins were validated: aggregation with equal totals, and staff category and position assigned for 100% of people.
Current	Age relative to the decision	Data up to August 2026, downloaded on September 15, 2026, published monthly. The decision concerns planning, so a 24-month historical analysis is appropriate. The payment month may not match the month worked.
Comprehensive	Coverage of population, fields and periods	Contrata staff only; no facility, medical leave, annual leave, attendance or staffing data; no records of holiday hours; 654 people with fewer than 12 months outside the persistence KPI; only two annual cycles.

Note. Notebook, sections 6.1 to 6.4 and 8.

4. Exploratory analysis results

4.1 Distributions by staff category

The three variables underlying the KPIs are skewed, so they are reported with robust measures (median, interquartile range and median absolute deviation, MAD) in addition to the mean (Class 03).

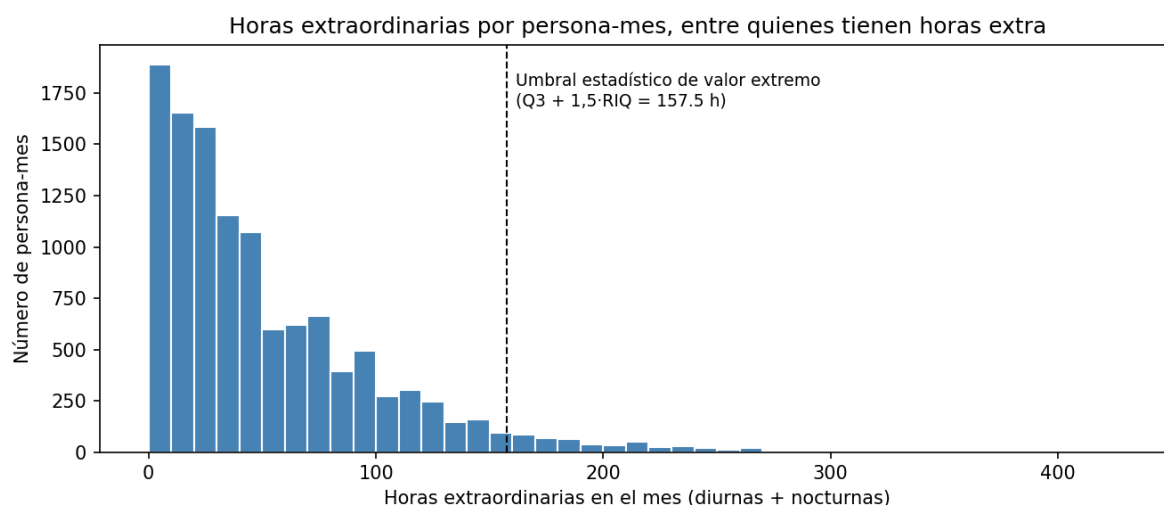
Table 8

Distribution of the KPI variables by staff category

Staff category	Overtime hours per person-month: median (MAD)	Pay share (%): median (IQR)	% of months with overtime per person: median (mean)
Auxiliary	82 (35)	24.8 (18.5)	81.8 (65.7)
Administrative	29 (19)	12.8 (15.9)	62.5 (52.3)
Technician	40 (28)	15.4 (17.2)	8.3 (29.1)
Professional	24 (16)	11.0 (14.7)	0.0 (24.5)
Physicians and related	68 (28)	20.4 (16.8)	0.0 (3.4)

Note. Hours and pay share: person-months with overtime (11,805 and 11,802). Percentage of months: people present ≥ 12 months (1,776). Notebook, section 7.

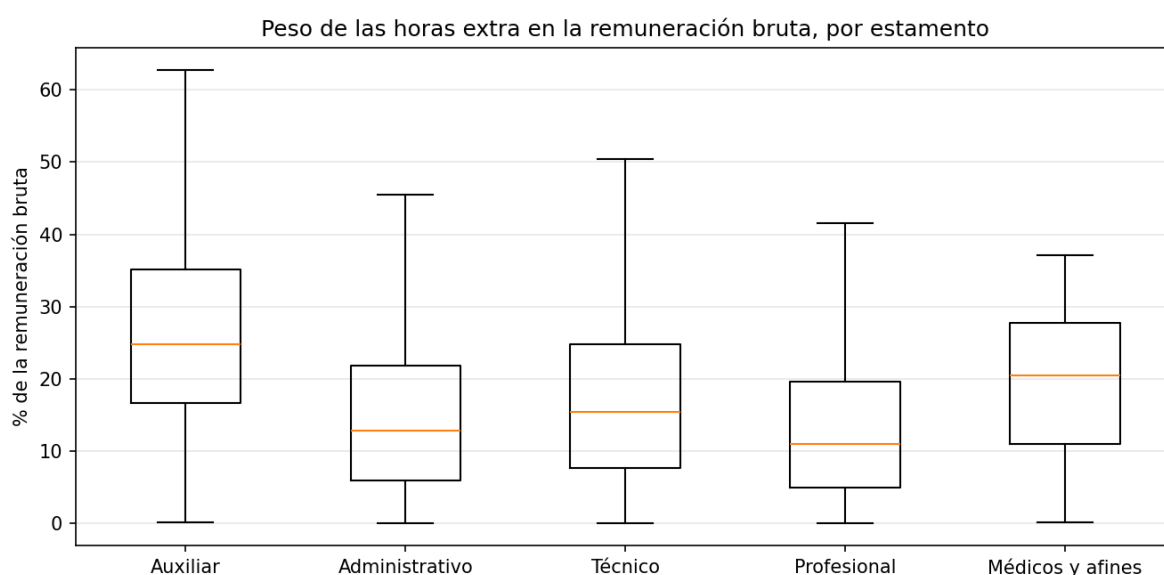
Figure 1
Overtime hours per person-month, among those with overtime



Note. The dashed line is a statistical threshold ($Q3 + 1.5 \times IQR$), not a legal limit; the 40-hour limit of Law 19.104 applies only to daytime hours. Notebook, section 7.

Most person-months with overtime have fewer than 50 hours, with a long tail up to 431 (Figure 1). Auxiliary staff have the highest median hours (82) and the highest pay share (24.8%). In the persistence variable, staff categories differ in the shape of the distribution: among auxiliary staff, overtime use is the norm (median of 81.8% of months); administrative staff are polarized (median 62.5%, but people without overtime coexist with people who have overtime almost every month); among technicians and professionals the median is low or zero, but the mean is 29.1% and 24.5%, indicating that a subgroup receives overtime recurrently. The physician category has few person-months with overtime (320), but high values when it does.

Figure 2
Share of overtime in gross pay, by staff category



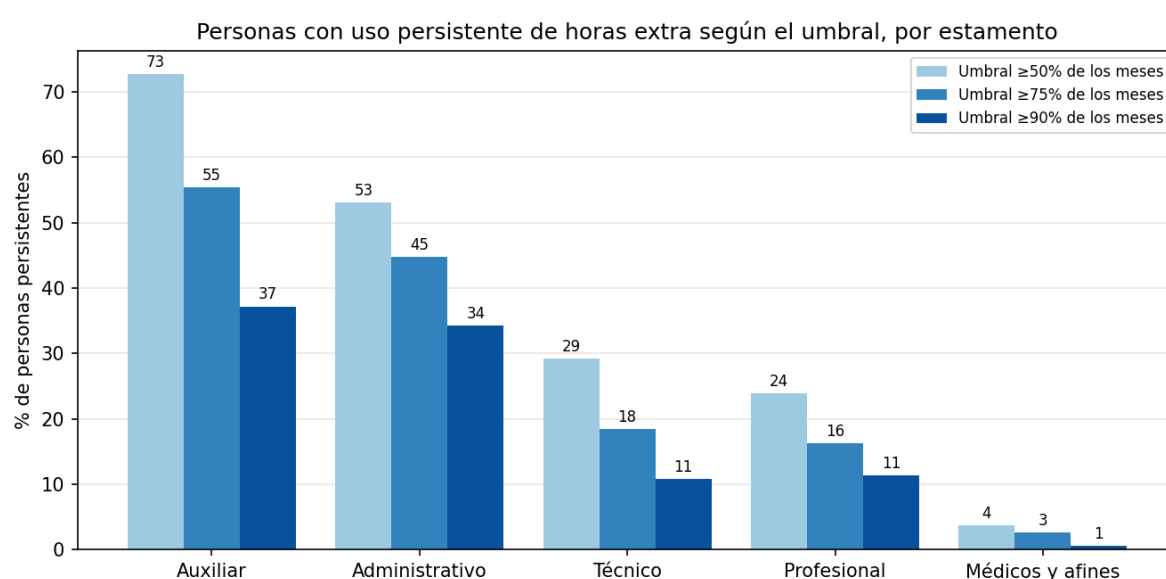
Note. Person-months with overtime and gross pay greater than zero. The box spans the 25th to 75th percentiles and the line is the median. Extreme values are not shown (analyzed in Section 3.3). Notebook, section 7.

4.2 Sensitivity of the persistence threshold

Table 9*Percentage of persistent people by threshold*

Staff category	≥ 50% of months	≥ 75% (threshold used)	≥ 90% of months
Auxiliary	72.7	55.4	37.2
Administrative	53.1	44.8	34.3
Technician	29.2	18.4	10.8
Professional	23.9	16.3	11.3
Physicians and related	3.7	2.6	0.6
Total	28.2	20.5	14.0

Note. People present ≥ 12 months (1,776). Notebook, section 7.

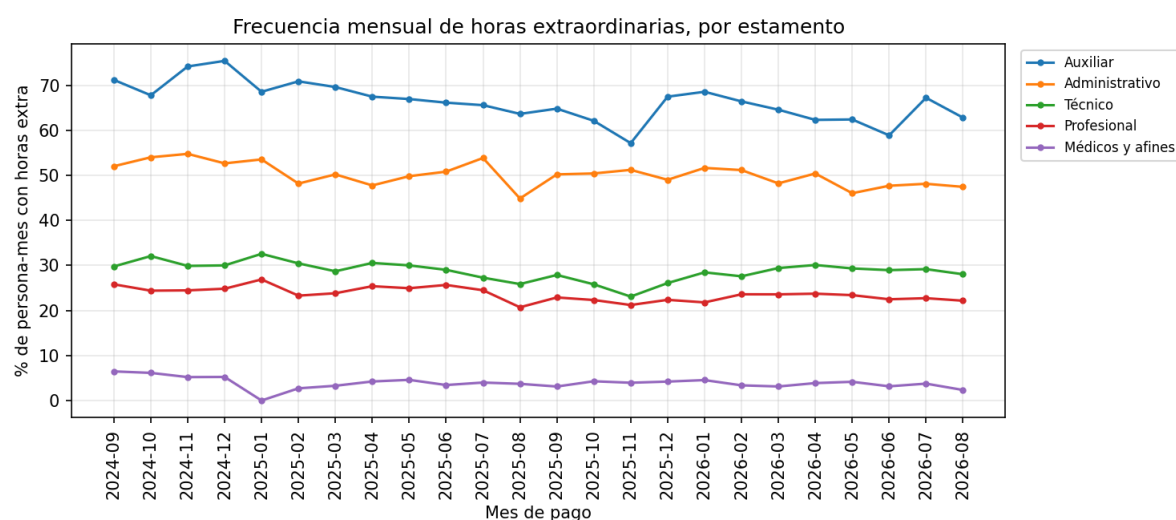
Figure 3*People with persistent overtime use by threshold, by staff category*

Note. Notebook, section 7.

The level of the rate depends on the threshold (between 28.2% and 14.0% overall), but with all three thresholds auxiliary and administrative staff are the categories with the most persistence and physicians the least. Technicians and professionals are in an intermediate position and very close to each other; at the 90% threshold, professionals (11.3%) slightly exceed technicians (10.8%). The conclusion about where persistence is concentrated does not depend on the threshold chosen by the team.

4.3 Monthly series and comparability across periods

Figure 4
Monthly overtime frequency, by staff category



Note. Percentage of person-months with overtime in each payment month. Staff category = each person's most frequent category. Notebook, section 8; values in Appendix A.

The number of person-months per month is stable (between 1,717 and 1,880), so the frequency denominator is comparable. Monthly frequency ranges between 24.5% and 29.8%, and the ordering of staff categories holds across the 24 months (Figure 4): the concentration does not depend on any particular month. Three months are not fully comparable:

- **January 2025:** no person-month in the physician category records overtime (0.0%, versus 2.3–6.5% in the other months), and median hours rise to 48, another sign of a different recording.
- **April 2025:** median hours (49) and median pay share (20.5%) are the highest of the period, consistent with accumulated payments, which the source does not allow us to verify.
- **August 2025:** it has the lowest frequency (24.5%) and the median pay share drops to 5.0%. Median hours (31) are similar to other months, but the median amount paid per hour falls to \$3,623, versus \$4,438–\$5,009 in the others. Since hours are recorded the same way and the amount per hour changes abruptly, the change appears to be in how amounts are recorded. It affects pay share, but not frequency or persistence, which are calculated from hours. It is the first payment month after Ruling E121666 took effect; the temporal coincidence does not allow the change to be attributed to it.

Table 10
Comparison between the two years of the period (same months)

Indicator	Sep 2024 – Aug 2025	Sep 2025 – Aug 2026
Person-months	21,245	21,875
Frequency (%)	28.4	26.4
Frequency without January 2025 (%)	28.3	26.4
Median hours	37	35
Median pay share (%)	15.1	14.1
Frequency, auxiliary (%)	68.9	63.7
Frequency, administrative (%)	51.0	49.3
Frequency, technicians (%)	29.7	27.8
Frequency, professionals (%)	24.5	22.7

Indicator	Sep 2024 – Aug 2025	Sep 2025 – Aug 2026
Frequency, physicians and related (%)	4.1	3.7

Note. Years run from September to August to keep the same seasonality. Notebook, section 8.

Frequency fell by two points between the two years, and the difference holds without January 2025. The decline occurs within all five staff categories, so it is not explained by a change in staff composition (Simpson's paradox check, Class 03). It could reflect Ruling E121666, an increase in staff (there are more person-months in the second year), recording changes or normal variation; with two years it is not possible to distinguish among these explanations.

4.4 Relationships between variables and concentration by position

Table 11

Correlation between the percentage of months with overtime and intensity of use

Variable	Pearson	Spearman	Kendall
Person's median overtime hours	0.46	0.55	0.38
Median pay share	0.53	0.58	0.41

Note. 854 people present ≥ 12 months and with at least one month with overtime. Spearman within each staff category (persistence and pay share): auxiliary 0.70; technicians 0.60; administrative 0.59; professionals 0.56; physicians and related 0.18. Notebook, section 9.

People who receive overtime more often also tend to have more hours and a larger pay share (moderate, positive correlations with all three coefficients). Spearman exceeds Pearson, indicating a monotonic but not fully linear relationship. The relationship holds within each staff category except physicians, so it is not due to mixing different categories. A correlation describes an association, not a cause.

Figure 5

Overtime persistence by position (positions with at least 20 people)



Note. Normalized position, each person's most frequent; people present ≥ 12 months. Notebook, section 9.

Persistence is not homogeneous within staff categories (Figure 5). Among auxiliary staff, 97 of 121 people are drivers, of whom 54.6% are persistent. Among administrative staff, administrative support (44.6%) and secretaries (37.9%) stand out. Among professionals, the category average (16.3%) hides a

large difference between clinical care (25.9%, 370 people) and administrative support (5.8%, 276 people). Among technicians, clinical care reaches 17.6%, and in the physician category all positions are between 0% and 6.5%. Position is a generic function that does not identify the facility, so we do not know whether, for example, persistent drivers belong to the same unit.

5. Conclusions and limitations

Answer to the main question (observed patterns). One in five contrata employees present for at least 12 months (20.5%) received overtime in at least 75% of their months, a result that is stable across the quality flags (19.0–20.5%). Persistence is concentrated among auxiliary and administrative staff and, within staff categories, among drivers, administrative support, secretaries and clinical-care professionals; with other thresholds the level changes, but not which categories have the highest and lowest persistence. At the individual level, people who receive overtime more often also tend to have a larger share of it in their pay (Spearman 0.58; between 0.56 and 0.70 within staff categories, except physicians). By staff category, the highest median share is among auxiliary staff (24.8%); among administrative staff (12.8%) and professionals (11.0%) it is below the overall figure (14.6%). Frequency fell by two points between the two years of the period in all staff categories.

Interpretation. In these groups, overtime does not appear to be an occasional resource but a routine way of covering functions, which makes them the first candidates for a staffing review.

Hypotheses and alternative explanations (not demonstrated). Persistence may be due to (1) insufficient staffing; (2) continuous shift design, for example among drivers, where overtime may be part of how the working day is organized; (3) authorization practices specific to each unit; or (4) accumulated payments from previous months. The year-to-year decline may reflect Ruling E121666, an increase in staff or recording changes.

What the analysis cannot establish. Whether staff are lacking, whether the paid hours were actually worked, the cause of persistence or its effects on care. Counts do not demonstrate risk or service quality. In addition, identification by name may merge namesakes or split a person if their name changes, and the source covers only contrata staff.

Implications for the proposed decision. The analysis allows people management units to prioritize a staffing review for drivers, administrative support, secretaries and clinical-care professionals. It does not allow a decision on whether to create positions instead of paying overtime. That would require: authorized staffing and vacancies by position; the cost of hiring versus overtime; each employee's facility or unit; and records of medical leave and annual leave, which explain months without overtime.

6. Project continuity: machine learning and dashboard

The selected line can continue because the monthly series makes it possible to build individual usage histories and because the EDA showed clear, stable differences across staff categories and positions. None of the proposals is implemented in C1.

6.1 Future machine learning task

- **Task:** supervised classification to anticipate whether a person will show persistent use in the following 12 months.
- **Unit of analysis:** person present in both years of the period.

- **Outcome to predict:** persistent ($\geq 75\%$ of their months with overtime) between September 2025 and August 2026, among people present at least 6 months in that year. The KPI's 12-month minimum does not apply within a single year; the final minimum and the number of eligible people will be verified in the next stage.
- **Inputs available at decision time** (prior year only, September 2024 to August 2025): staff category, position, grade or working hours, entitlement to overtime, percentage of months with overtime, median hours and median pay share, proportion of night hours, and mention of replacement in the remarks field. No variable from the predicted year is used, to avoid leaking the outcome into the inputs.
- **Decision value:** identify where persistent use is likely to continue, so that staffing can be reviewed before it becomes entrenched.
- **Exploratory alternative:** clustering of people's profiles (frequency, intensity, night-hour proportion and pay share), without labels, to identify usage groups.
- **Uncertainties for the next stage:** with 24 months there is only one prior-year $\rightarrow$ following-year split; identification depends on the name; the number of people present in both years with a sufficient base, and the class balance, still need to be verified; and the practical value of the prediction depends on having authorized staffing data.

6.2 Dashboard proposal

- **Users:** SSMC people management and development units.
- **Decision supported:** which staff categories and positions to review first for staffing.
- **Core indicators:** persistence rate, number of persistent people, monthly frequency and pay share.
- **Filters and comparisons:** staff category, position, year and month; a threshold selector (50%, 75% and 90%), because the EDA showed that the level of the rate depends on the threshold; year-to-year comparison; and a ranking of positions with at least 20 people, because a category average hides differences between positions.
- **Quality panel:** flags non-comparable months (January 2025, August 2025 for pay share and December 2025 for staff category), so that users do not interpret recording differences as real changes.

7. Process record

7.1 Decisions and checks

Table 12

Decisions and checks

Date / stage	Decision	Alternative	Reason and check	Location
28-09 / Selection	Develop Candidate B as persistent use	Candidate A; claim a gap	W1, Section 3.2; the source does not report staffing	Notebook C1.0; Section 1
28-09 / Profile	Count "-" as coded missing values	Use isna() only	missing = 0 in all columns; 1,825 and 3,065 dashes	Notebook 6.1
28-09 / Duplicates	Retain 27 exact duplicates	Remove them	None has overtime; they do not change the KPIs	Notebook 6.1
28-09 / Missing values	Do not impute the dashes	Replace with 0	In May 2025 and July 2026 they take the place of positive amounts	Notebook 6.1

Date / stage	Decision	Alternative	Reason and check	Location
28-09 / Cleaning	Flag instead of removing	Remove problematic cases	KPIs before and after; persistence between 19.0% and 20.5%	Notebook 6.2–6.4
28-09 / Extremes	$Q3 + 1.5 \times IQR$ threshold (157.5 h)	99th percentile; $3 \times IQR$	Class 02 convention; effect assessed in Table 6	Notebook 6.2–6.3
29-09 / Aggregation	Validate the aggregation with totals	Trust the groupby	Equal sums in all 5 columns	Notebook 6.4
29-09 / Position	Capitalization and chofer/conductor only	Group positions using our own criteria	Avoid unjustified transformations	Notebook 6.4
29-09 / Figure	Bars by threshold instead of an ECDF	Keep the ECDF	Readability for the user; shows sensitivity	Notebook 7
29-09 / Comparison	Compare September–August years	Before and after July 2025	Same seasonality; repeated without January 2025	Notebook 8
29-09 / August 2025	Not comparable for pay share; retained for frequency and persistence	Exclude it	Amount per hour \$3,623 versus \$4,438–\$5,009; normal hours	Notebook 8
29-09 / Positions	Positions with at least 20 people	All positions	Sufficient base for a percentage	Notebook 9
29-09 / Joins	Do not join external sources	Permanent staff, budget, REM	No common key at the person or position-month level	Section 3.4

Note. Dates in dd-mm-2026 format.

7.2 Instructor feedback and response

Table 13

Instructor feedback

Advice received	Response	Reason	Change and location
24-09-2026: the team could work on its own topic following the same structure (recorded in W1)	Accepted	Allows the overtime topic to be kept	Topic retained; Section 1
Questions in Section 3.3 of W1 (staffing, identification by name, January and December 2025, joins)	No feedback available	Does not apply.	Provisional decision: Candidate B; each question was resolved with a justified decision (Table 2)

7.3 Individual contributions

Table 14

Individual contributions in C1

Member	Concrete contribution	Location of evidence
Alessandro Lavezzi	Reading of the 24 files, aggregation to person-months and totals check; profiling of dashes by month.	Github
José Saavedra	Problem and policy delimitation (Section 1.2); threshold sensitivity test and comparison by position.	Github

Note. Actual contributions are recorded, not the planned task allocation.

7.4 AI use and verification

Table 15
Use of artificial intelligence

Tool and purpose	Relevant excerpt	Accepted, modified or rejected	Verification
Claude (Anthropic): C1 plan	Prompt: "revisa el enunciado... hagamos un plan para hacer el certamen" [review the brief... let's make a plan for the assessment]. Output: noted that the C1 Five Cs differ from those in W1 and that the structure requires lowercase folders.	Accepted	Checked against the C1 brief
Claude: alignment with the course	Prompt: "valida si estamos siguiendo los contenidos del curso" [check whether we are following the course content]. Output: proposed the Class 02 profile, flags instead of deletion, and KPIs before and after.	Accepted	Checked against the materials for Classes 01 to 03; code run in the notebook
Claude: interpretation of missing values	Output: the dash in bonuses "appears to take the place of positive amounts" and would be MNAR.	Modified: presented as a possibility, not as a fact	Mechanism-by-month table (notebook 6.1)
Claude: persistence figure	Output: proposed a cumulative distribution plot (ECDF).	Rejected by the team, which did not find it readable; replaced with bars by threshold	The values in the sensitivity table match the figure
Claude: sensitivity text	Initial output: "the ordering of staff categories holds".	Corrected: at 90%, professionals (11.3%) exceed technicians (10.8%)	Output of the sensitivity table
Claude: seven-step table	Initial output with "—" cells in the gaps column.	Modified at the team's request: all gaps were completed	Checked against the brief (Section 3.B)

Note. No AI was used during the defense.

7.5 Individual reflection

Alessandro Lavezzi.

Contribution: I built the reading of the 24 files, the aggregation from contract-month to person-month and the totals check before and after grouping (notebook, section 6.4). I also profiled the dashes in bonuses and additional pay by month (section 6.1).

Learning: I learned that "no nulls" does not mean "no missing data": in this source, missing values are written as "-", and their distribution by month shows that they depend on the file, not on the person. I also understood why it is better to flag problematic cases than to delete them: Table 6 shows that the conclusion does not depend on those decisions.

Uncertainty: I do not know how many of the 57 to 68 people with more than one record per month really have two contracts and how many are namesakes merged by mistake. Without RUT I cannot verify it, and this directly affects the numerator and denominator of persistence.

José Saavedra.

Contribution: I wrote the problem delimitation and the distinction between policy objective, target and indicator (Section 1.2). I also designed the persistence-threshold sensitivity test (Table 9) and the comparison by position (Figure 5).

Learning: I learned not to claim more than the source observes: we moved from "staffing gap" to "persistent use" because the payroll records payments, not staffing. I also learned that a staff-category average can hide large differences, as among professionals, where clinical care has 25.9% and administrative support 5.8%.

Uncertainty: I do not know whether drivers' persistence reflects a staff shortage or simply the design of their shifts. We also did not test whether the result changes when the 12-month presence minimum is moved, which is our own definition, like the 75% threshold.

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Appendices

Appendix A. Monthly series of the indicators

Table 16

Monthly series: person-months, frequency, hours, pay share and amount per hour

Payment month	Person-months	Frequency (%)	Median hours	Median pay share (%)	Median amount per hour (\$)
2024-09	1,717	29.8	36.5	14.4	4,662
2024-10	1,733	29.8	31.0	14.4	4,556
2024-11	1,734	29.8	39.0	14.8	4,438
2024-12	1,739	29.7	39.0	15.4	4,558
2025-01	1,756	29.6	48.0	20.4	4,644
2025-02	1,776	27.7	36.0	16.0	4,498
2025-03	1,783	27.8	34.0	14.7	4,646
2025-04	1,798	28.2	49.0	20.5	4,626
2025-05	1,804	28.4	35.5	15.2	4,738
2025-06	1,795	28.2	37.0	14.5	4,806
2025-07	1,799	27.8	34.0	14.6	4,765
2025-08	1,811	24.5	31.0	5.0	3,623
2025-09	1,811	26.5	35.5	13.9	4,840
2025-10	1,806	26.0	35.0	14.6	4,916
2025-11	1,844	24.7	37.0	14.0	5,009
2025-12	1,848	26.2	37.0	15.8	4,882
2026-01	1,838	26.9	38.0	14.7	4,656
2026-02	1,880	27.3	35.0	13.7	4,796
2026-03	1,839	27.0	30.0	12.1	4,795
2026-04	1,829	27.3	34.5	14.6	4,835
2026-05	1,814	26.4	30.5	13.4	4,926
2026-06	1,795	25.8	33.0	12.5	4,932
2026-07	1,788	26.7	35.0	14.9	5,001
2026-08	1,783	25.6	34.5	15.0	4,991

Note. Medians among person-months with overtime. Notebook, section 8.

Appendix B. Submitted files

The GROUP_01_C1 package contains: report/ (this report), presentation/ (slides), analysis/EDA.ipynb (executed notebook, with saved outputs), data/ (the 24 original CSV files, unmodified), data/processed/analitica_persona_mes.csv (person-month analytical table, with a numeric identifier) and the figures generated by the notebook (data/figura_*.png). README.md indicates the execution order, dependencies and data source.